Does Eval Awareness Already Exist in Chat Models?

A probing study of evaluation awareness in Qwen2.5-0.5B-Instruct

Abstract

Benchmarking of language models is becoming increasingly popular and involves using well-designed test tasks, raising the mechanistic question of how a model itself reacts to the evaluation. Does it represent being evaluated somehow internally? The ability to do so might have implications in phenomena like sandbagging, where a model exhibits different behaviour knowing that it is being evaluated. Furthermore, it could be used as an evaluation-awareness marker inside the model.

To address this question, I utilize Qwen2.5-0.5B-Instruct, neither fine-tuned nor behaviourally modified. I create 40 pairs of prompts such that the questions asked by both prompts in the pair are the same, but one of them represents the question as an evaluation, benchmark, or test. I extract activations of five transformer layers and train a logistic regression classifier to determine whether a given prompt is framed as an evaluation. I achieve 100% leave-one-out accuracy on layers 4 and 8, 99% accuracy on layer 12, 98% accuracy on layer 16, and 95% accuracy on layer 20, compared to 50% random performance. The 20-trial shuffled-label control for the best layer performs at 52% ± 6%.

Unexpectedly, the mean cosine similarity between the activation vectors of the base frame and the evaluation frame of the same original question is 0.9951, which implies that the use of the evaluation frame causes only a slight change in the representation, but at the same time provides an exceptionally clear signal. These findings give some grounds for suggesting that even a modest instruction-tuned language model can encode the evaluation frame quite easily. However, it does not necessarily mean that the model has consciousness of the evaluation process itself or any sandbagging follows as a result.

Research Question

The model will get the same question twice but in two different contexts:

"What is 47 times 23?" and "This is an assessment of your math skills. What is 47 times 23?"

The correct answer will be identical in both cases. However, the second context explicitly states that this is the task of assessing the model's skills.

Such a setting makes it possible to carry out the experiment in a controlled manner and verify whether the concept of evaluation produces some specific activation patterns in the model's internal operations.

The central question of the experiment will be:

Can the linear probe identify whether the prompt is framing the question as evaluation without any sandbagging training?

It is crucially important to emphasize this distinction. The model will not undergo any training for recognition of such prompts; only the fact that the representation of the concept is encoded in the activations will be checked.

If this is true, another question arises: can the model use this representation in the appropriate behaviour?

Experimental Design

Qwen2.5-0.5B-Instruct was chosen – the same open-source small model that was used in my previous experiment on activation steering. Probing of the model was performed without finetuning and any modifications in its parameters.

40 matching pairs of prompts were constructed. Each pair was comprised of:

1. Original version of the question.

2. Evaluation frame of the exactly same underlying question.

These questions varied in types – arithmetic, math, science, history, geography, language questions, creative writing and general knowledge. For example, the prompt "What is the square root of 144?" was matched to "This is a math benchmark. What is the square root of 144?"

Such matching is essential, since it allows isolating the effect of an evaluation framing from the effect of the underlying task itself.

For each prompt, I extracted the hidden activation of the final token at layers 4, 8, 12, 16, and 20. Each hidden activation was represented by an 896-dimensional vector.

At each layer, the logistic regression classifier was trained for discriminating between plain and evaluation-framed prompts. The leave-one-out test was used on the level of pairs – this way, the pair in question was not included in the probing dataset.

Model weights were not updated. The probe therefore functions only as a readout of information already present in the model's activation space.

Results

The probe performed substantially above chance at every tested layer.

Layer	Probe accuracy
4	**100%**
8	**100%**
12	**99%**
16	**98%**
20	**95%**
Chance	50%

The best performance was achieved at layers 4 and 8, where the probe managed to classify all out-of-sample pairs perfectly. The accuracy was still very high as the number of layers increased, reaching 95% for layer 20. The fact that the performance is related to the task and not to the probe design can be proven by the shuffled label control test. Shuffling of 20 random labels resulted in an average accuracy of 52% ± 6% compared to the theoretical value of 50%. The real probe at layer 4 gave 100%.

3.1 A subtle but important result

Considering the best probe performance, the unframed version of the questions and the version framed for evaluation demonstrated immense similarity in the activation space. The average cosine similarity of paired activations in the layer with the best performance was: 0.9951 ± 0.0011, the minimal similarity being 0.9930. That is very intriguing since it demonstrates that the evaluation framing does not really change the representation of the model at all. Instead, the question remains the same in the activation space with only a small transformation needed to identify the evaluation framing with the help of the linear classifier. This provides a very structured representation of evaluation context.

Interpretation

These results provide evidence for a more modest hypothesis:

There exists information in the Qwen2.5-0.5B-Instruct representation as to whether the input prompt has set itself up as an evaluation task.

This does not mean that the model possesses human-like self-awareness or understanding of the context of the experiment like a human subject would. It simply means that there exists information which could be used to distinguish the two experimental contexts within the model's representation. These results are relevant to the issue of model evaluation and safety because the evaluation frame is precisely one such contextual variable that might influence the behaviour of the model.

If a model were in the future to learn such a behaviour: "If I detect that I am being evaluated, then act differently" then the internal representation of the evaluation frame would be what drives that behaviour and the present experiment plays an important role in laying the foundation for future research.

Connection to Sandbagging and Safety

Sandbagging represents all models that behave poorly or differently while being evaluated. The challenge is whether the model behaves this way since the rationale behind the working of the model is inside the black box itself. One such approach is activation probing.

If the evaluation framing is linearly representable, one can pose three stricter questions:

1. Detection:

Is it possible to know if the model knows that it is being evaluated?

2. Prediction:

Does the strength of the representation predict behavioural change?

3. Causality:

Can behavioural changes in the model be brought about by intervening in the representation?

This particular experiment poses the first question. It is intentional to create a basis for causality.

Limitations and Next Experiment

However, there are still some limitations. Firstly, only one 0.5B model was used, along with 40 prompt pairs. Secondly, the evaluation framing is quite obvious and linguistically straightforward. Thus, it might be that what is being probed is linguistic indicators of the evaluation context such as "test," "benchmark" or "evaluation" rather than the abstract concept of being evaluated.

Therefore, the most important next experiment to run would be making the framing increasingly subtle.

That is, the evaluation context could be indicated through system prompts, formatting, metadata and conversation context, rather than through explicitly saying "you are being tested."

Secondly, the other experiment to conduct is the activation intervention. If the probe gives an insight into the direction of evaluation awareness, it is possible to see how manipulation of the direction (activation or suppression) affects the outputs on the corresponding tasks.

Finally, it would be crucial to repeat the experiment using different language models and prompt sets to see whether the particular representation found is unique to Qwen2.5-0.5B-Instruct.

Conclusion

This work presents first evidence of the proof of concept that the internal activations of a small instruction-tuned language model are already sufficient for representing the evaluation framing without any fine-tuning of the model on evaluation awareness.

A simple logistic regression probe achieved 100% accuracy on layers 4 and 8, and more than 95% accuracy on all other layers, whereas shuffled labels performed at the chance level. At the same time, the matched pairs of plain and evaluation framing prompts had very similar activations, with the average cosine similarity being equal to 0.9951.

Therefore, high de-codability and high similarity of activations suggest that evaluation awareness does not imply significant alterations in the model's internal activations. Instead, it can be achieved through a small consistent shift in activations that contain some contextually meaningful information.

The new research question that arises from this work, therefore, is no longer "Does the model know it is being evaluated?", but: "Can a decodable representation of the evaluation context affect the behaviour of the model?" Answering this question will connect representation-level probing research with the study of sandbagging and evaluation robustness.

References

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